



REPUBLIC OF GHANA
OFFICE OF THE HEAD OF THE CIVIL SERVICE
INTERNAL AUDIT DEPARTMENT

Product Requirements Document

IAD Digital Platform — the official web presence and operational backbone of internal audit across the Ghana Civil Service

DOCUMENT	Product Requirements Document (PRD)
PRODUCT	IAD Digital Platform — https://iad.ohcsghana.org
PREPARED BY	OHCS IT Department
PREPARED FOR	Internal Audit Department (IAD)
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Document Control

Title	Product Requirements Document — IAD Digital Platform
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Author	OHCS IT Department
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Live platform	https://iad.ohcsghana.org
Source repository	github.com/ghwmlite-dotcom/IAD (branch: main)

Revision History

VERSION	DATE	CHANGE
1.0	May 2026	First issue — platform as deployed through Tier 2 (Knowledge Hub)
1.1	May 2026	Annex A: vendor valuation replaced with internal operationalization budget
1.2	May 2026	Annex A: budget reduced to GHS 35,000 via free tiers and absorption
1.3	May 2026	Annex A: budget restricted to technical line items only
1.4	May 2026	Annex A: optional AI module added (GHS 50,000 total if adopted)
1.5	May 2026	Closing note reframed for submission to IAD management

Approval

Director of Internal Audit Product Owner — Date	Director, RSIMD Delivering Unit — Date	Head of Civil Service Noted — Date
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Contents

1	Executive Summary	3
2	Background & Problem Statement	3
3	Goals & Success Metrics	4
4	Personas & Access Model	4
5	Shipped Scope (v1.0)	5
6	Architecture	6
7	Non-Functional Requirements	6
8	Known Limitations	7
9	Roadmap	7
10	Risks & Dependencies	7
A	Annex A — Operationalization Budget	8

1 Executive Summary

The IAD Digital Platform is the official web presence and operational backbone of the Internal Audit Department of Ghana's Civil Service. It serves two missions at once:

1. **Public mission** — position the IAD as a transparent, modern, globally credible audit institution: publishing audit findings, the state of internal audit across the service, and a verifiable professional registry of Internal Audit Class officers.
2. **Operational mission** — give the IAD and every MDA Internal Audit Unit a single digital headquarters: audit universe management, annual planning, engagement tracking, findings & recommendations follow-up, and an authoritative Knowledge Hub.

Institutional context. The IAD is the headquarters of internal audit in the Ghana Civil Service: every Ministry, Department and Agency (MDA) has its own Internal Audit Unit with its own head, and all of them answer to the Director of Internal Audit. The platform is architected around that reality — it is built for many units, not one department.

2 Background & Problem Statement

Before this platform:

- The IAD had no dedicated digital identity; it existed as a sub-section of the OHCS website with no operational tooling.
- Audit findings and recommendations lived in spreadsheets and paper files; there was no public accountability surface and no consolidated view of implementation status.
- Internal Audit Class officers had no verifiable professional identity — credentials (FCCA, ACCA, IIA, CITG) could not be checked by employers, donors or the public.

Officers' qualifications lived on paper CVs and in personnel files: an employer posting or promoting an officer, a development partner funding a PFM programme and seeking assurance that the audit units involved were competently staffed, or a journalist, researcher or citizen — none had an independent way to confirm that an officer held what they claimed.
- Standards, manuals and templates circulated informally; MDA units had no single authoritative source for IAD guidance.
- Whistleblowing channels lacked provable anonymity guarantees.

3 Goals & Success Metrics

GOAL	METRIC
Transparency leadership	Press/academic citations of the public findings tracker; donor references
Operational adoption	MDA liaison accounts active; findings with management responses on file
Professional registry	Officers with verified digital profiles; certificate verifications served
Knowledge centralisation	Knowledge Hub documents published; downloads served
Platform excellence	Sub-second global TTFB; 95+ Lighthouse accessibility; zero critical security findings

4 Personas & Access Model

PERSONA	DESCRIPTION	ACCESS
Public citizen / journalist / researcher	Consumes transparency data, verifies auditors, downloads publications, submits whistleblower reports	Public pages, no login
MDA audit unit staff (liaison)	Manages their MDA's audit universe, plans, engagements and findings	Portal — role: liaison
IAD auditor / manager	Cross-MDA engagement work, findings review, working papers	Portal — roles: auditor, manager
Director of Internal Audit	Full oversight, publishing decisions, administrative functions	Portal + Admin — role: director
Platform administrator	Users, registry, content, Knowledge Hub, site configuration	Admin — role: admin

Authentication. Passwordless magic-link email sign-in with role-based sessions — a signed-in device is remembered for 30 days (sliding renewal, 30-day hard cap) before re-authentication is required.

5 Shipped Scope (as of v1.0)

5.1 Public Website

- Fully bespoke IAD brand and design system: hero slider designed around the internal-audit mission; leadership section featuring the Director of Internal Audit and the OHCS Internal Audit Unit head; news, events, services, contact.
- Bilingual chrome (EN/FR); body content in English.
- SEO/AEO pass: prerendered content pages, JSON-LD structured data, sitemap, and paginated listing pages built to scale to hundreds of entries.
- Mobile-responsive across the entire platform (audited and fixed); newsletter subscription.

5.2 Public Findings Tracker (/track)

Public, searchable tracker of audit findings and recommendations with implementation status — the transparency flagship.

5.3 "State of Internal Audit" Transparency Page (/transparency)

Live dashboard: audit-plan coverage, aggregate risk heat map, resolution-rate trends. Aggregate queries only, cached at the edge.

5.4 Internal Audit Class Registry (/registry, /verify)

Verified professional profiles for Internal Audit Class officers (credentials, CPD), QR-verifiable digital certificates, and a public verification endpoint. Paginated and prerendered for search visibility at scale.

5.5 Audit Operations Portal (/portal)

Audit universe (MDAs, auditable entities), risk register, annual planning, engagements with working-paper uploads, findings & recommendations tracker, notifications, and per-MDA liaison views.

5.6 Knowledge Hub Tier 2 — shipped

- **Public** (/publications): searchable, paginated library of manuals, templates, standards, circulars, guidelines, reports, forms and policies with versioned, downloadable files.
- **Portal** (/portal/knowledge): adds MDA-only restricted documents.
- **Admin**: full publishing workflow — upload (PDF/DOCX/XLSX ≤ 25 MB), metadata, draft/publish, versioning with change notes, audience control.

5.7 Admin Console (/admin)

Registry management, users & roles, Knowledge Hub publishing, submissions (whistleblower/feedback) review pipeline, leadership, news & events, site configuration, and an audit log of every administrative mutation. AI-training features disabled by directive.

5.8 Whistleblowing & Submissions

Public reporting form with sealed reference numbers and a review pipeline in admin. Provable-anonymity hardening is scheduled in Phase 3 (see §9).

6 Architecture

LAYER	TECHNOLOGY
Frontend	Next.js 16 (static export), React 19, Tailwind CSS v4
Edge hosting	Cloudflare Pages — custom domain iad.ohcsghana.org with SSL
Backend	Cloudflare Pages Functions (serverless, same-origin /api/*)
Database	Cloudflare D1 (SQLite at the edge) — 15 migrations applied
File storage	Cloudflare R2 — working papers and Knowledge Hub documents
Auth	Magic-link email, role-based sessions
Quality	423 automated tests (Vitest), static build gate, documented API contract

Key architectural decisions

- **Static export + edge functions** — the whole site is served from Cloudflare's edge network for sub-second global loads; all dynamic behaviour goes through same-origin APIs.
- **Contract-first backend** — docs/API-CONTRACT.md defines the entire API surface.
- **Audit-everything** — every administrative mutation writes to an audit log.
- **Fail-soft prerendering** — build-time data snapshots degrade gracefully, so builds never break on API unavailability.

7 Non-Functional Requirements

- **Performance** — sub-second TTFB worldwide via the edge network; paginated lists (12/page) so pages stay fast as data grows.
- **Security** — role-gated endpoints, session hardening, magic-byte file-type validation on uploads, Content-Disposition sanitisation, no client-side secrets.
- **Privacy** — whistleblower submissions designed for anonymity; Phase 3 adds formal anonymity hardening and a published protocol.
- **Reliability** — idempotent seed/migration tooling; demonstration data clearly separated from production roster data.
- **Accessibility** — semantic HTML, ARIA labelling, keyboard-operable modals and toggles; formal WCAG pass scheduled in Phase 3.

8 Known Limitations (v1.0)

1. Magic-link login requires the production email-service (Resend) API key to be set as a deployment secret.
2. Seeded demonstration data (MDAs, findings, registry officers) awaits replacement with the real roster — bulk CSV import is available as a follow-on.
3. News/events admin screens exist; final wiring of those content types is optional polish.
4. French translation covers site chrome; full body-content translation is Tier 3.

9 Roadmap

PHASE	SCOPE	STATUS
Phase 1	Live backend: D1 schema, API, magic-link auth, admin CMS	Shipped
Phase 2	Audit Operations Platform — universe, risk, planning, engagements, findings, liaison portal	Shipped
Tier 1	Public findings tracker, transparency page, IAC registry & CPD	Shipped
Tier 2	Knowledge Hub	Shipped
Phase 3	Hardening: security review, whistleblower provable-anonymity certification, accessibility pass, performance, launch checklist	Next
Tier 2 (rem.)	AI-assisted audit copilot (internal; gated on security review and real data)	Planned
Tier 3	Full bilingual content, annual "State of Accountability" microsite, performance & accessibility as brand	Planned

10 Risks & Dependencies

- **Data ownership** — the real officer roster and findings data must come from IAD; the platform's value compounds only when real data flows.
- **Email deliverability** — magic-link authentication depends on the email service key and DNS records remaining valid.
- **Continuity** — repository, hosting account and database live under the OHCS Google account; credentials handover should be formally documented.

A Annex A — Operationalization Budget

Internal delivery. The platform is developed in-house by the OHCS IT Department for the Internal Audit Department. No external development fees apply. The budget below is restricted to **technical line items only** — the engineering work and running costs required to operationalize and keep the platform — so that every figure is directly traceable to the platform itself. Training, launch and documentation activities are absorbed internally by OHCS IT and carry no cost here.

A. One-off technical costs

#	ITEM	WHAT IT COVERS	GHS
1	Cloud infrastructure provisioning	Cloudflare Pages/Functions environments, D1 database setup, R2 storage, DNS & SSL configuration, deployment secrets	6,000
2	Data migration & digitisation	Engineering work to move the real Internal Audit Class officer roster and findings & recommendations records into the platform database, with verification	8,000
3	Independent security assessment	External penetration test of the platform and anonymity review of the whistleblowing channel before go-live	9,000
4	Backup, monitoring & disaster recovery	Automated database backups, uptime monitoring, alerting and recovery runbooks	4,000
5	Email authentication infrastructure	Magic-link email service integration and deliverability configuration (SPF/DKIM)	3,000
6	Contingency ($\approx 10\%$)		2,600
Subtotal — one-off			32,600

B. Recurrent annual costs

#	ITEM	WHAT IT COVERS	GHS/YR
1	Transactional email service	Magic-link authentication emails (entry tier)	1,800
2	Domain & DNS	ohcsghana.org renewal share	600
Subtotal — recurrent			2,400

Cost absorptions that make the GHS 35,000 possible: Cloudflare Pages, D1 and R2 run on **free tiers** at launch volumes (GHS 0); platform maintenance, training, launch and documentation are **absorbed by OHCS IT** as part of normal duties (GHS 0).

D. Optional enhancement — AI-Assisted Audit Capabilities (not mandatory)

Entirely optional. This module may be adopted at any later stage without rework — the platform's architecture already anticipates it (see §9, Tier 2: AI-Assisted Audit Copilot). It is presented separately so the core GHS 35,000 ask stands on its own.

#	ITEM	WHAT IT COVERS	GHS
1	AI copilot integration	Cloudflare Workers AI binding, working-paper drafting assistance and sampling support inside the audit portal	8,000
2	Anomaly-flagging pipeline	Upload-and-analyse flow for financial data, flagging outliers for auditor review	4,000
3	AI governance & security review	Data-confidentiality controls, prompt isolation, and audit logging of all AI-assisted actions	3,000
Subtotal — optional AI module			15,000

Recurrent impact: negligible at launch — inference runs on the Workers AI free allocation; usage-based costs are reviewed annually.

E. Summary

	GHS
Core platform — year one (mandatory)	35,000
Optional AI module (one-off, if adopted)	15,000
Project total with AI module	50,000
Subsequent years (core platform)	2,400 / yr

Conclusion. The year-one operationalization budget of **GHS 35,000** for the IAD Digital Platform — with an optional AI-assisted audit enhancement of GHS 15,000, taking the total to GHS 50,000 only if adopted — is respectfully submitted to IAD management for consideration and approval. Every line is a technical cost attached to the platform: infrastructure, data, security, resilience and email. All non-technical activities are absorbed in-house by the OHCS IT Department.

